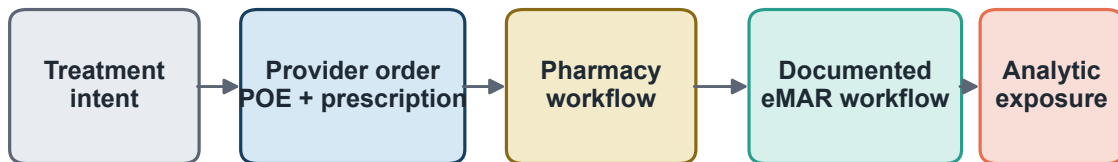
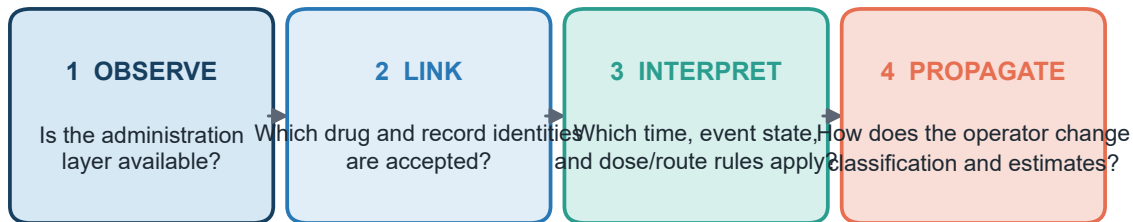


Medication exposure is a joint provenance operator

A source becomes exposure only after identity, time, semantics, and required metadata are resolved



Exposure operator = observable source × identity × time × event semantics × dose



Discordant provenance cells transmit operator choices to cohort composition and estimates